Metode Numerice @ CS-UPB

Echipa MN CS - 2015:

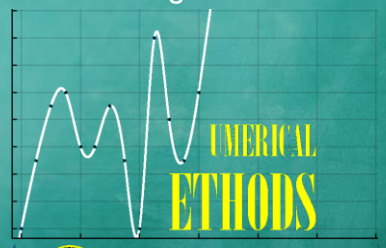
Profesori:

- Pantelimon George Popescu - Seria CA;
- Florin Pop - Seria CB, CC;

Asistenti:

PG Popescu, Clementin Cercel, Sorin Ciolofan, Bogdan Tiganoaia, Diana Udrea, Alexandru Tifrea, David Iancu, Bogdan Cristian Marchis, Radu Poenaru, Madalina Grosu, Andrei Vlad Postoaca, Radu Constantinescu, Roxana Istrate, Madalina Hristache, Adelina Vidovici;

Transformari ortogonale
Polinoame ortogonale



Transformari ortogonale

Produsul scalar $\langle x, y\rangle = y^* x$ se numeste produs euclidian pentru doi vectori x,y\in C^n.

Norma euclidiana pentru un vector complex x este valoarea reala $||x||_2 = \sqrt{\langle x, x\rangle}$.

- $\langle ax1 + bx2, y\rangle = a\langle x1, y\rangle$+b$\langle x2, y\rangle$;
- $\langle x, ay1 + by2\rangle = \bar{a}\langle x, y1\rangle$+$\bar{b}\langle x, y2\rangle$;
- CS ineq: $|\langle x, y\rangle| \leq ||x||_2 ||y||_2$;

Transformari ortogonale

- Vectorii v1,v2,…,vn sunt ortogonali daca $vi^T vj = 0$ pentru orice i≠j;
- Vectorii v1,v2,…,vn sunt ortonormali daca $vi^T vj = 0$ pentru orice i≠j si $vi^T vi = 1$ pt orice i;

Orice lista de vectori ortonormali este linear indepedenta si orice lista de vectori ortogonali, diferiti de vectorul zero este linear independenta.

Transformari ortogonale

Din orice lista de vectori linear independenti S pot obtine o lista de vectori ortonormali cu aceeasi span.

$$\operatorname{span}(S) = \left\{ \sum_{i=1}^{k} \lambda_i v_i \mid k \in \mathbb{N}, v_i \in S, \lambda_i \in \mathbf{K} \right\}$$

Procesul se poate face in multe moduri dar exista un proces sistematizat, Gram-Schmitd (Jørgen Pedersen Gram((1850 - 1916) - mat. danez) si Erhard Schmidt((1876 -1959) - mat. german))

Transformari ortogonale

Daca v1,v2,…,vn linear independenti, prin GS obtin o lista z1,z2,…,zn de vectori ortonormali cu span{v1,v2,…,vk}=span{z1,z2,…,zk} pt orice k=1…n; Vectori zi se calculeaza astfel:

y1=x1; z1=y1/||y1||;

y2=x2-<x2,z1>z1; z2=y2/||y2||;

…

$$y_k = x_k - \langle x_k, z_{k-1}\rangle z_{k-1} - \langle x_k, z_{k-2}\rangle z_{k-2} - \cdots - \langle x_k, z_1\rangle z_1$$

zk=yk/||yk||;

Transformari ortogonale

Daca consideram Z=[z1 … zn] si X=[x1 … xn] atunci procesul GS ne obtine o factorizare X=ZR unde R este nesingulara si superior triunghiulara.

Daca x1,…,xk sunt ortonormali si x1,…,xk,xk+1,…,xn sunt linear independenti atunci procesul GS obtine x1,…,xk,zk+1,…,zn ortonormali.

Transformari ortogonale

Daca H este o matrice nxn, ortogonala atunci

- H^-1=H^t;
- det(H)=+-1;
- $\|Hx\|_2=\|x\|_2$ - conserva norma euclidiana a unui vector;
- $\|H\|_2=1$ - norma euclidiana subordonata 1;
- $\|H.A\|_2=\|A\|_2$ - conserva norma euclidiana a unei matrici;

Transformari ortogonale

Daca H este o matrice nxn, ortogonala atunci

- Vectorii coloane sunt ortogonali;
- Vectorii linii sunt ortogonali;

Daca U,V sunt unitare(ortogonale in R) atunci si UV este unitara(ortogonala in R).

Transformari ortogonale

Householder (Alston Scott Householder (1904 -1993) matemtician american)

- Fie w din R^n astfel incat w^tw=1, matricea nxn H=I_n-2ww^t se numeste transformare Householder(matrice Householder).
- Pentru a evita conditia w^tw=1 se poate alege H=I_n-2ww^t/w^tw.
- Matricea H este simetrica si ortogonala(unitara si hermitica).

Transformari ortogonale

Householder

- Matricea H are valori propri 1 si -1, defapt are o valoare proprie -1, prima si restu 1;
- det(H)=-1;
- Orice x vector real poate fi transfromat de o matrice Householder reala in orice y vector real, cu ||x||=||y||(norma euclidiana);

Transformari ortogonale

Householder

- Ne propunem sa aducm o matrice simetrica A la o forma tridiagonala.
- Metoda consta la fiecare pas in a gasi matricea Hk care sa faca zero sub a doua diagonala.
- Initial se cauta H1 astfel incat matricea A2=H1AH1 sa aiba zero in prima coloana din linia 3, si pentru ca inmultim cu H1 la dreapta se obtine 0 pe prima linie de la coloana 3.

Transformari ortogonale

Householder

$$\alpha = -\mathrm{sgn}(a_{k+1,k}^{(k)}) \left(\sum_{j=k+1}^{n} (a_{jk}^{(k)})^2 \right)^{1/2},$$

pentru k=1,2,…,n-2

$$r = \left(\frac{1}{2}\alpha^2 - \frac{1}{2}\alpha a_{k+1,k}^{(k)} \right)^{1/2},$$

$$w_1^{(k)} = w_2^{(k)} = \ldots = w_k^{(k)} = 0,$$

$$w_{k+1}^{(k)} = \frac{a_{k+1,k}^{(k)} - \alpha}{2r},$$

$$w_j^{(k)} = \frac{a_{jk}^{(k)}}{2r}, \quad j = k+2, k+3, \ldots, n,$$

$$P^{(k)} = I - 2\mathbf{w}^{(k)} \cdot (\mathbf{w}^{(k)})^t,$$

$$A^{(k+1)} = P^{(k)} A^{(k)} P^{(k)}$$

Transformari ortogonale

Householder

$$A^{(k+1)} = \begin{bmatrix} a_{11}^{(k+1)} & a_{12}^{(k+1)} & 0 & \cdots & 0 \\ a_{21}^{(k+1)} & & & & \\ 0 & & & & 0 \\ & & a_{k+1,k}^{(k+1)} & a_{k+1,k+1}^{(k+1)} & a_{k+1,k+2}^{(k+1)} \cdots a_{k+1,n}^{(k+1)} \\ & & 0 & & \\ 0 & \cdots & 0 & a_{n,k+1}^{(k+1)} & \cdots a_{nn}^{(k+1)} \end{bmatrix}.$$

se continua pana cand A=An-1 devine tridiagonala

$$A^{(n-1)} = P^{(n-2)} P^{(n-3)} \ldots P^{(1)} A P^{(1)} \ldots P^{(n-3)} P^{(n-2)}$$

Transformari ortogonale

Householder

Cand se doreste transformarea unei matrici intr-o matrice triunghiular superioara.

Fie A o matrice mxn, atunci exista H mxm ortogonala astfel incat HA=R, unde R este o matrice mxn superior triunghiulara.

Transformari ortogonale

Householder

$$H_p = I_m - 2\frac{v_p v_p^T}{v_p^T v_p}$$

$v_p=[0\ldots0\ v_{pp}\ldots v_{mp}]^T$

$$\sigma_p = \mathrm{sign}(a_{pp}) \sqrt{\sum_{i=p}^{m} a_{ip}^2}$$

$v_{pp}=a_{pp}+\sigma_p$, $v_{ip}=a_{ip}$, i>p.

Ap+1=HpAp, A1=A;

HA=Hn…H2H1A;

Transformari ortogonale

Exemplu Householder(triunghiular):

Alegeti ce vreti voi!

Transformari ortogonale

Givens (James Wallace Givens(1910-1993))

- O matrice G care difera de matricea unitate in cel mult patru elemente, care sunt de forma gii=gjj=cosθ si gij=-gji=sinθ, pentru un anume θ si i!=j, se numerste matrice de rotatie.
- GA difera de A doar pe randurie i si j;
- AG difera de A doar pe coloanele i si j;
- Orice matrice de rotatie G este ortogonala;

Transformari ortogonale

Givens

Pentru i<j,

Gi,j=G(θ,i,j)=

$$\begin{bmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & \cos\theta & & -\sin\theta & & \\ & & & & 1 & & & \\ & & & & & \ddots & & \\ & & & & & & 1 & \\ & & & \sin\theta & & \cos\theta & & \\ & & & & & & & 1 \\ & & & & & & & & \ddots \\ & & & & & & & & & 1 \end{bmatrix}$$

Transformari ortogonale

Givens

- Matricea Givens, G se mai numeste matrice de rotatie plana.
- Gij este ortogonala pentru orice pereche i<j si orice θ din [0,2pi).
- G(θ,i,j)^-1=G(- θ,I,j).

Transformari ortogonale

Givens

Cand se doreste ca prin transformari Givens sa obtinem o matrice superior triunghiulara, atunci alegerea lui θ se face astfel incat GAji=0, i!=j, adica din ga_ji=aiisin θji +ajicos θji=0, de unde rezulta tgθji=-aji/aii, ρ=sqrt(aii^2+aji^2) iar cosθij=aii/ρ si sinθij=-aji/ρ

Transformari ortogonale

Givens

Deci la fiecare pas se elimina elementele de sub diagonala principala, aplicand cate o transformare Givens(matrice diferita de fiecare data) pentru fiecare element!(la Householder aplicam o transfromare si eliminam toate elementele de sub pivot)

$$G = \underbrace{G_{n-1,n}}_{G_{n-1}} \cdot \underbrace{G_{n-2,n} \cdot G_{n-2,n-1}}_{G_{n-2}} \cdots \underbrace{G_{1n} \cdot G_{1,n-1} \cdots G_{12}}_{G_1}$$

Transformari ortogonale

Exemplu Givens:

Alegeti voi!

Transformari ortogonale

Factorizari QR Householder & Givens

Dupa cum am observat in ambele cazuri cand am aplicat transformari ortogonale(H si G) asupra unei matrici A, am obtinut o matrice R superior triunghiulara, adica:

Hn...H2H1A= R_H sau Gn...G2G1A=R_G

Unde matricile de transformare Hi si Gi sunt ortogonale, deci inversele lor sunt transpusele.

Transformari ortogonale
Factorizari QR Householder & Givens

Daca inmultim cu inversa lui H, respectiv inversa lui G in stanga, identitatile vor deveni

A=(Hn…H2H1)$^{-1}R_H$ sau A=(Gn…G2G1)$^{-1}R_G$

Echivalent cu

A=$(H1)^{-T}(H2)^{-T}\dots(Hn)^{-T}R_H$ sau

A=$(G1)^{-T}(G2)^{-T}\dots(Gn)^{-T}R_G$

Si daca notam produsul inverselor transformatelor corespunzator, obtinem

Transformari ortogonale
Factorizari QR Householder & Givens

A=$Q_H R_H$ sau A=$Q_G R_G$

unde Q sunt matrici ortogonale, iar R matrici superiro triunghiulare. Deci transformarile ortogonale pot fi privite si ca factorizari QR.

Transformari ortogonale
Householder despre Givens(o nota interesanta)

Unitary Triangularization of a Nonsymmetric Matrix*

ALSTON S. HOUSEHOLDER

Oak Ridge National Laboratory, Oak Ridge, Tennessee

A method for the inversion of a nonsymmetric matrix, due to J. W. Givens, has been in use at Oak Ridge National Laboratory and has proved to be highly stable numerically but to require a rather large number of arithmetic operations, including a total of $n(n-1)/2$ square roots. Strictly, the method achieves the triangularization of the matrix, after which any standard method may be employed for inverting the triangle. The triangular form is brought about by means of a sequence of $n(n-1)/2$ plane rotations, whose product is an orthogonal matrix. Each rotation requires the extraction of a square root. The advantage in using the method lies in the fact that an orthogonal matrix is perfectly conditioned. Hence the condition of the matrix cannot deteriorate through successive transformations. In fact, if one defines the condition number of a matrix A to be [1]

$$\gamma(A) = \| A \| \, \| A^{-1} \|,$$

Transformari ortogonale
Householder despre Givens(o nota interesanta)

The purpose of the present note is to point out that the same result can be obtained with fewer arithmetic operations, and, in particular, for inverting a square matrix of order n, at most $2(n-1)$ square roots are required, instead of $n(n-1)/2$. For $n > 4$, this is a saving of $(n-4)(n-1)/4$ square roots.

After $n-1$ steps, at most, the matrix A is triangularized:

$$U = U_{n-1}U_{n-2}\cdots U_1, \qquad UA = R.$$

Transformari ortogonale
Householder despre Givens(o nota interesanta)

If the matrix

$$A = (a_1, a_2, \cdots, a_n)$$

is scaled at the outset so that $\| a_i \| \leq 1$ for every i, then all elements remain within range throughout the triangularization, since a unitary transformation leaves the Euclidean norm invariant. Hence no scaling problems arise in the actual triangularization. Moreover, when R^{-1} is formed, if this is similarly scaled no further scaling is required in multiplying by the matrices U_i to form A^{-1}. Only in forming R^{-1} itself may intermediate scaling be required.

Polinoame ortogonale

Daca vi este un sir de vectori linear independenti atunci exista un sir de vectori ui ortonormati, cu span{vi}=span{ui}(construiti de exemplu cu GS).

La fel se intampla si la polinoame, inlocuim sirul vi cu polinoamele {1,x,x^2,…,x^n} din care putem obtine un sir de polinoame ortogonale {p0,p1,…,pn}.

Cand vorbim de ortogonalitate ne referim la un anumit produs scalar.

O familie de polinoame ortogonale se defineste in mod unic in raport cu intervalul de definitie [a,b] si o functie pondere w(x).

Polinoame ortogonale

Cand vorbim de ortogonalitate ne referim la un anumit produs scalar. De obicei, pentru functii se considera urmatoarele produse scalare

$$\langle g,h\rangle = \int_a^b g(x)h(x)w(x)\,dx,$$

$$\langle g,h\rangle = \sum_{i=1}^{n} g(x_i)h(x_i)w(x_i),$$

Polinoame ortogonale

Anumite produse scalare satisfac relatia de simetrie, adica <xf,g>=<f,xg> si in acest caz sirul polinoamelor ortogonale se deduce din relatiile

$$p_0(x)=1,\quad p_1(x)=x-\alpha_0$$

$$p_{k+1}(x)=(x-\alpha_k)p_k(x)-\beta_k p_{k-1}(x),\qquad k=1:n-1$$

$$\alpha_k=\frac{\langle xp_k,\ p_k\rangle}{\|p_k\|^2},\qquad k=0:n-1,$$

$$\beta_k=\frac{\|p_k\|^2}{\|p_{k-1}\|^2},\qquad k=1:n-1.$$

Polinoame ortogonale

Pentru un polinom

$$p_n(x)=a_{nn}x^n+a_{n-1,n}x^{n-1}+a_{0n}$$

care este ortonormat, avem relatia

$$\frac{a_{n,n}}{a_{n+1,n+1}}\cdot p_{n+1}(x)+\left(\frac{a_{n-1,n}}{a_{n,n}}-\frac{a_{n,n+1}}{a_{n+1,n+1}}-x\right)\cdot p_n(x)+\frac{a_{n-1,n-1}}{a_{n,n}}\cdot p_{n-1}(x)=0$$

iar daca este doar ortogonal avem relatia

$$\frac{a_{n,n}}{a_{n+1,n+1}}\cdot p_{n+1}(x)+\left(\frac{a_{n-1,n}}{a_{n,n}}-\frac{a_{n,n+1}}{a_{n+1,n+1}}-x\right)\cdot p_n(x)+\frac{a_{n-1,n-1}}{a_{n,n}}\cdot\frac{\|p_n\|^2}{\|p_{n-1}\|^2}\,p_{n-1}(x)=0$$

Polinoame ortogonale

Polinoame des utilizate

Cebâşev	$T_{n+1}-2xT_n+T_{n-1}=0,$	$T_0=1,\ T_1=x$
Legendre	$(n+1)L_{n+1}-(2n+1)xL_n+nL_{n-1}=0,$	$L_0=1,\ L_1=x$
Laguerre	$G_{n+1}-(2n+1-x)G_n+n^2G_{n-1}=0,$	$G_0=1,\ G_1=1-x$
Hermite	$H_{n+1}-2xH_n+2nH_{n-1}=0,$	$H_0=1,\ H_1=2x$

Pafnuty Lvovich Chebyshev ((1821-1894) mat. rus)
Adrien-Marie Legendre ((1752-1833) mat. francez)
Edmond Nicolas Laguerre ((1834-1886) mat. francez)
Charles Hermite ((1822-1901) mat. francez)

Polinoame ortogonale

Proprietati Polinoame ortogonale

- Radacini reale distincte situate in [a,b];
- Mentin propr. de ortogonalitate cu orice polinom(neortogonal) de grad mai mic;
- Radacinile polinomului Pn determina intervalele de separare a zerourilor pentru polinomul Pn+1;

Polinoame ortogonale

Proprietati Polinoame ortogonale

- Minimul integralei \int_a^b w(x)qn^2(x)dx pentru orice polinom de grad n, este realizat de polinomul ortogonal Pn(x), definit in mod unic in raport cu intervalui [a,b] si ponderea w(x);

Bibliografie Transformari ortogonale / Polinoame ortogonale

- Richard L. Burden, J. Douglas Faires, Numerical Analysis, 9th edition, Brooks/Cole, Cengage Learning, 2011;
- Roger A. Horn, Charles R. Johnson, Matrix Analysis, 2th edition, Cambridge, 2013;
- Valeriu Iorga, Boris Jora, *Metode Numerice*, Editura Albastră, 2005;

